

Kenya Certificate of Secondary Education (K.C.S.E)

449/1

DRAWING AND DESIGN

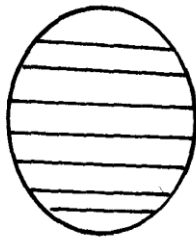
PAPER 1

MARKING SCHEME

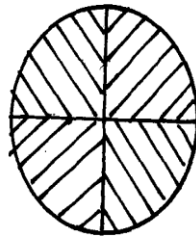
1.
 - i) Compressive strength – the ability to withstand squeezing load.
 - ii) Toughness – The ability of a material to withstand impact load or hammering load.
 - iii) Ductility – The process where the area of cross-section is reduced gradually while the length is increased in tension without rupture.
 - iv) Malleability – the ability of a material to extend permanently in all directions without rupture under compressive loads.
2.

Thermosetting plastics – They undergo irreversible chemical change when heated.

Thermoplastics – Always soften when heated and harden again on cooling.
- 3.



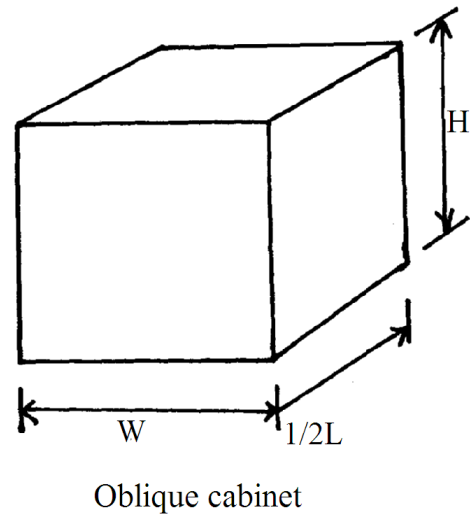
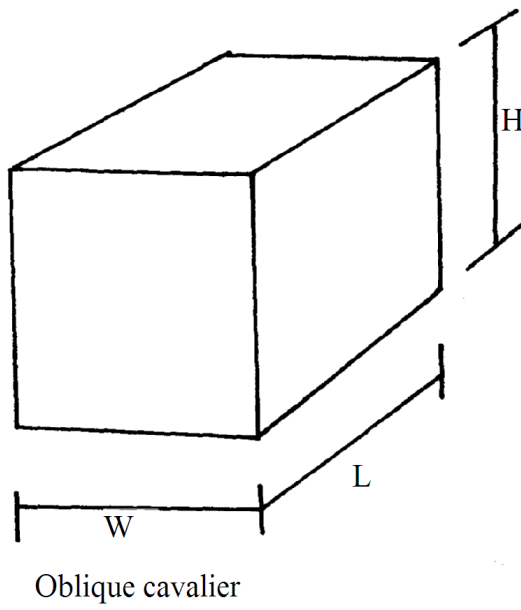
Plain



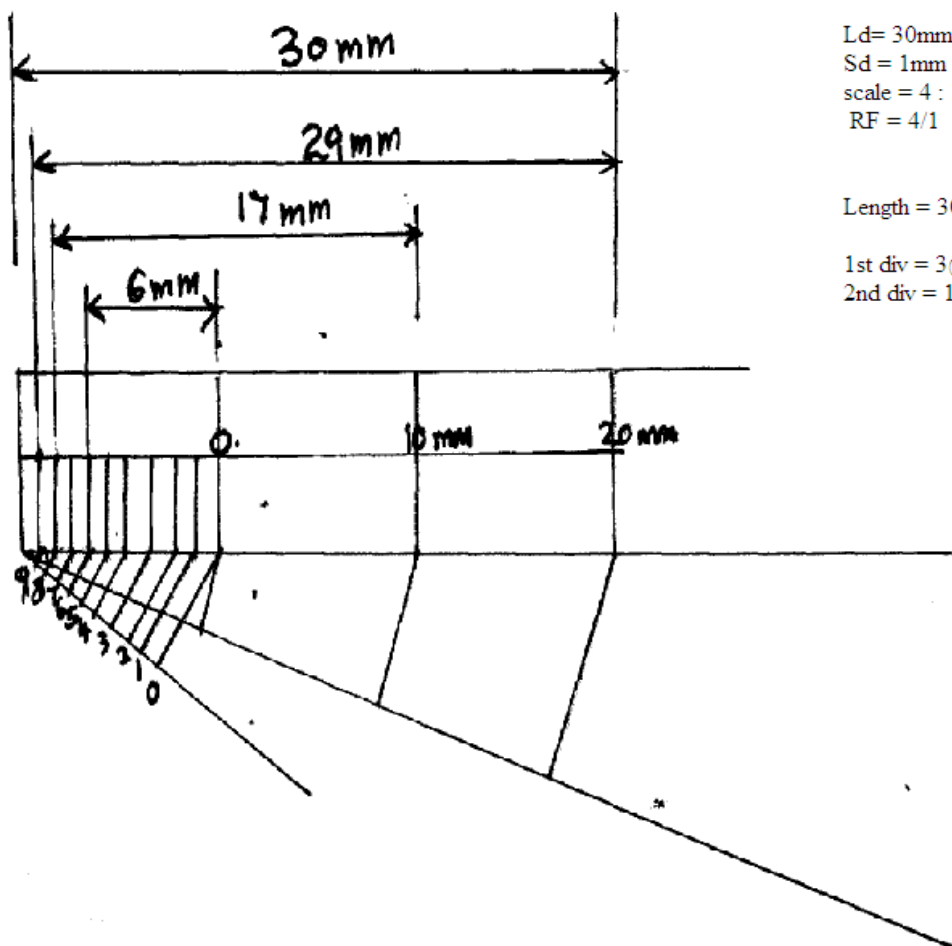
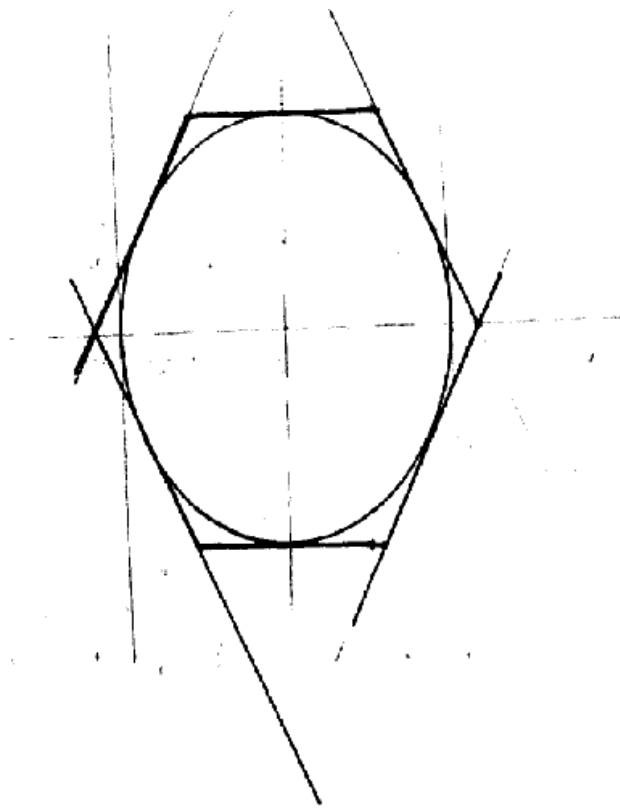
Quarter sawing

4.
 - i) Air seasoning
Timber is stacked in open-sided sheds which protect it from the sun and rain, but which permit the air to circulate freely. It is very slow process but gives best results.
 - ii) Kiln seasoning
The timber is stacked then hot air of controlled humidity and temperature is circulated through the stack. The hot air dries the timber while steam prevents it from splitting and warping.
5.
 - a)
 - i) Ferrous metal – metal which contain iron
 - ii) Non-ferrous metal – metal which do not contain iron e.g aluminium copper

b)



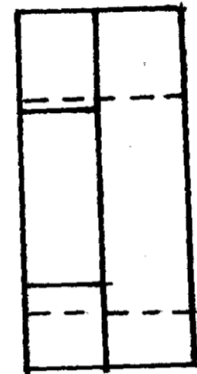
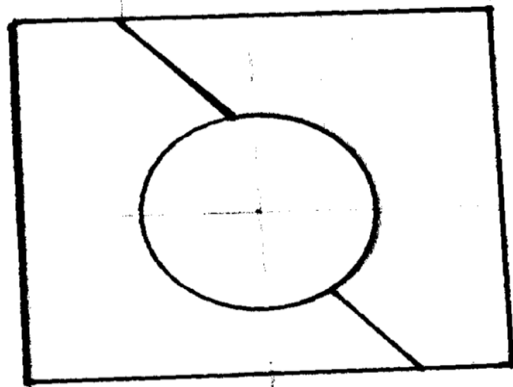
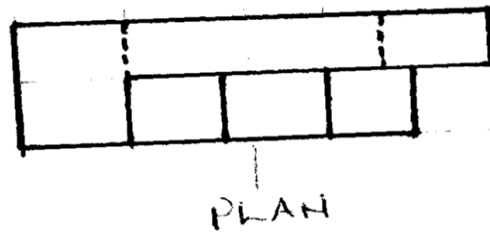
Actual measurements indicated on the drawing



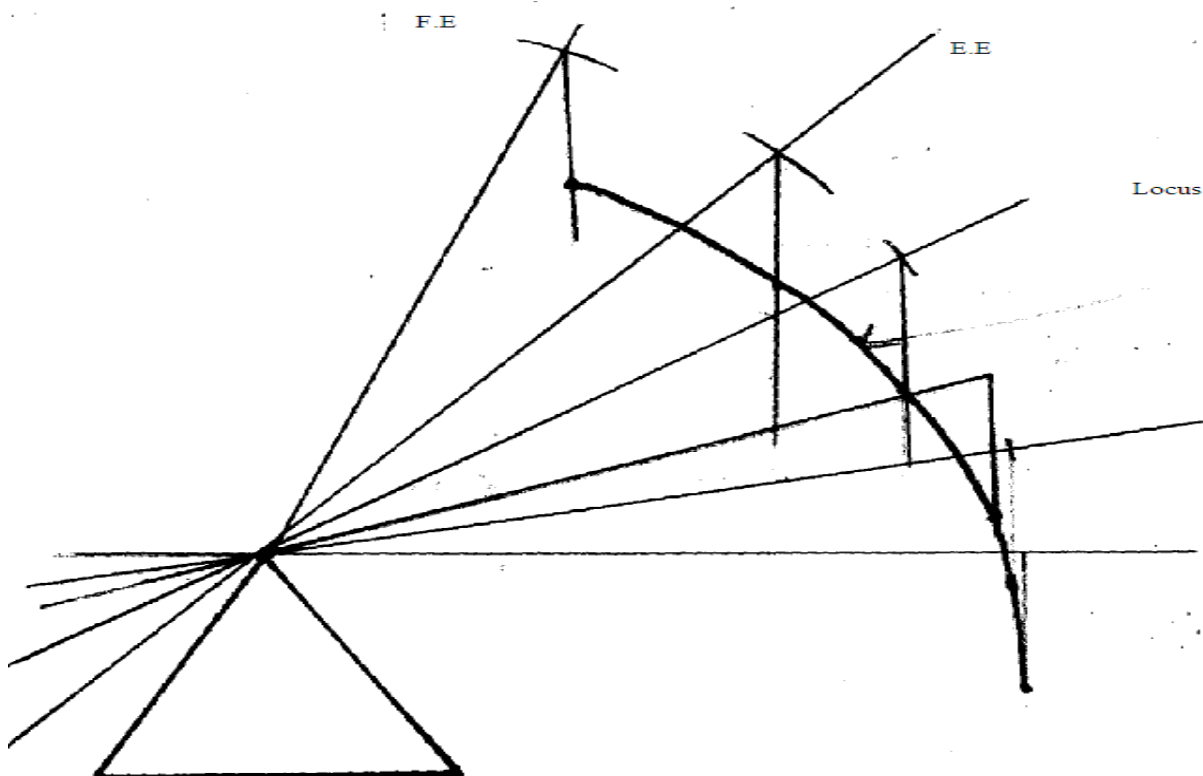
Ld = 30mm
Sd = 1mm
scale = 4 : 1
RF = 4/1

Length = 30mm x 4/1 = 120mm

1st div = 3@10mm
2nd div = 10@1mm



10.



F.E – 1MK

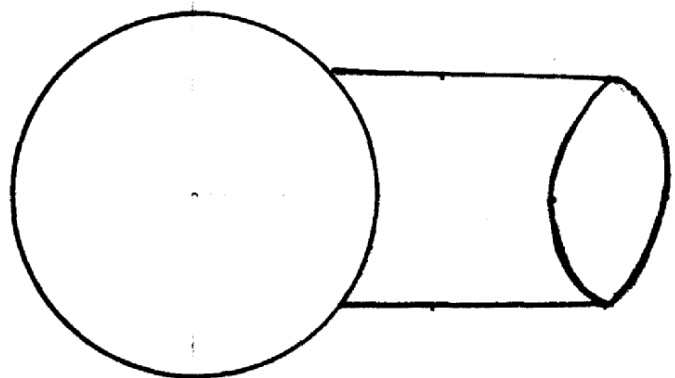
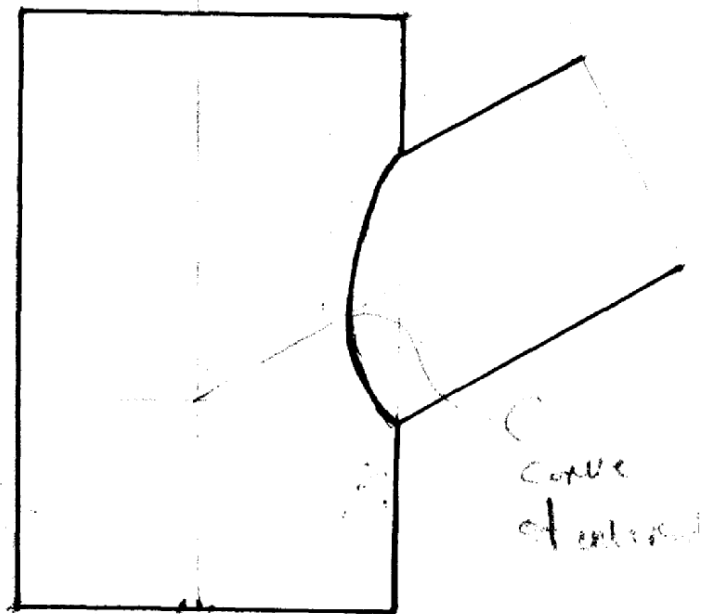
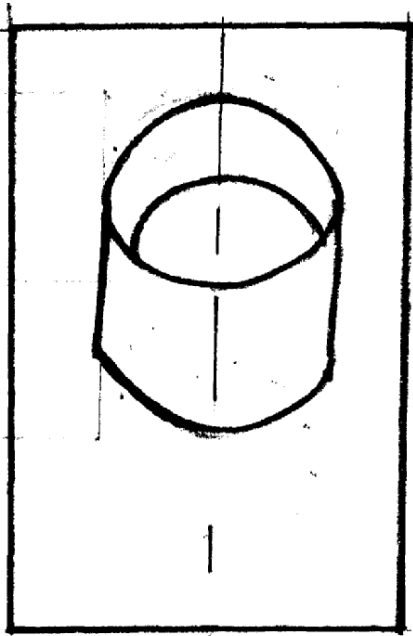
Plan – 3mks

E.E – 4mks

T.S – 2mks

Auxiliary view – 4mks

13.



14.

